

Task 2 – Motor Control with PWM & Obstacle Detection

1. AIM:

To control the speed of a DC motor using Pulse Width Modulation (PWM) and automatically stop the motor when an obstacle is detected using an HC-SR04 ultrasonic sensor.

2. GOAL:

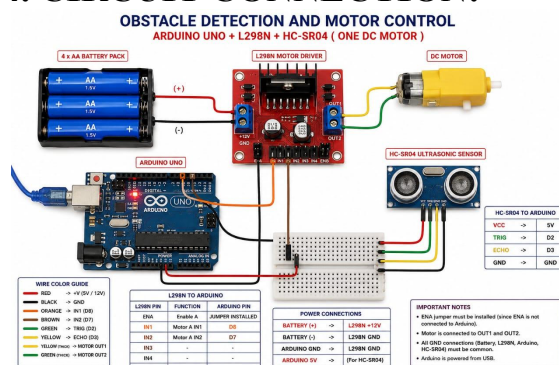
The goal of this task is to:

- Control a DC motor using Arduino UNO.
- Use PWM to control the motor speed.
- Measure the distance of an obstacle using the HC-SR04 ultrasonic sensor.
- Automatically stop the motor when an obstacle is detected within the specified distance.
- Demonstrate the relationship between obstacle distance and motor response.

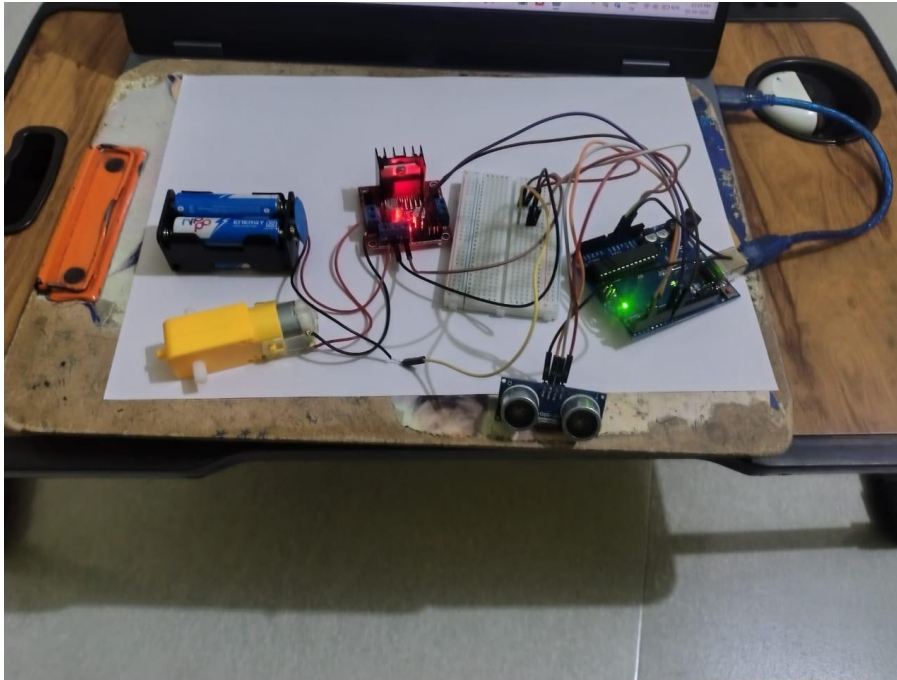
3. COMPONENTS REQUIRED:

S.No	Components	Qty
1	Arduino UNO	1
2	HC-SR04 Ultrasonic Sensor	1
3	L298N Motor Driver Module	1
4	DC Gear Motor	1
5	4 × AA Battery Holder	1
6	AA Batteries	4
7	Jumper Wires	As required
8	USB Cable	1

4. CIRCUIT CONNECTION:



5. COMPLETE CIRCUIT CONNECTION:



6. ARDUINO PROGRAM:

```
// TASK 2 - PWM Motor Control with Obstacle Detection

// HC-SR04 Pins
const int trigPin = 6;
const int echoPin = 7;

// L298N Motor Driver Pins
const int ENA = 5; // PWM Pin
const int IN1 = 8;
const int IN2 = 9;

// Settings
const int obstacleDistance = 15; // cm
const int motorSpeed = 180; // PWM value (0-255)

long duration;
float distance;

void setup() {

  Serial.begin(9600);

  // Ultrasonic sensor
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  // Motor driver
  pinMode(ENA, OUTPUT);
```

```

pinMode(IN1, OUTPUT);
pinMode(IN2, OUTPUT);

// Initially stop motor
analogWrite(ENA, 0);
digitalWrite(IN1, LOW);
digitalWrite(IN2, LOW);

Serial.println("TASK 2: PWM Motor Control");
}

void loop() {

// Trigger ultrasonic sensor
digitalWrite(trigPin, LOW);
delayMicroseconds(2);

digitalWrite(trigPin, HIGH);
delayMicroseconds(10);

digitalWrite(trigPin, LOW);

// Read echo signal
duration = pulseIn(echoPin, HIGH, 30000);

// Calculate distance
distance = duration * 0.0343 / 2;

// Print distance
Serial.print("Distance: ");
Serial.print(distance);
Serial.println(" cm");

// Obstacle detection
if (distance > 0 && distance <= obstacleDistance) {

// Stop motor
analogWrite(ENA, 0);
digitalWrite(IN1, LOW);
digitalWrite(IN2, LOW);

Serial.println("Obstacle Detected!");
Serial.println("Motor Stopped");

} else {

// Run motor
digitalWrite(IN1, HIGH);
digitalWrite(IN2, LOW);

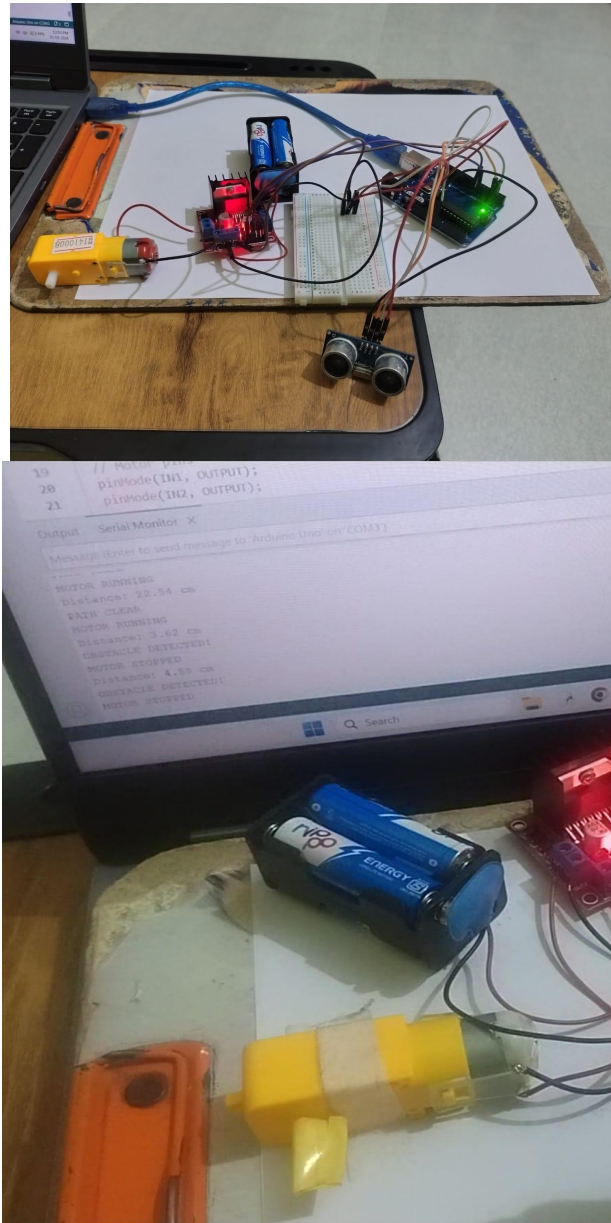
// PWM speed control
analogWrite(ENA, motorSpeed);

Serial.println("Path Clear");
Serial.println("Motor Running");
}
}

```

```
Serial.println("-----");  
  
delay(500);  
}
```

7. OUTPUT:



8. TASK 2 DEMO VIDEO:

https://drive.google.com/file/d/1nSF7Ihvfuwz_0iRVUrBM016ElV1Jbr7/view?usp=drivesdk

9. RESULT:

The PWM motor control and obstacle detection system was successfully implemented using an Arduino UNO, HC-SR04 ultrasonic sensor, L298N motor driver, and DC gear motor.

The ultrasonic sensor successfully measured the distance of obstacles. The Arduino controlled the motor speed using PWM. When an obstacle was detected within 15 cm, the motor automatically stopped. When the obstacle was outside the specified safety distance, the motor continued to run.

Thus, the objective of controlling the DC motor using PWM and automatically stopping it during obstacle detection was successfully achieved.